

# CURRICULUM VITAE

## PERSONAL INFO

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First Name: Vasiliki  
Last Name: Stoumpou  
E-mail: [vasstou@mit.edu](mailto:vasstou@mit.edu)

## EDUCATION

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- **PhD candidate at Operations Research Center,**  
**Massachusetts Institute of Technology** **09/2022 – 2027**  
GPA: 5.0/5.0  
Advisor: Prof. Dimitris Bertsimas  
Research: Working on Machine Learning, with applications in Healthcare.
- **Integrated Masters (5 years degree), National Technical University of Athens, Greece**  
**School of Electrical and Computer Engineering** **09/2016 – 07/2021**  
Grade: 9.59 / 10 (**ranked 3rd** among the graduates of 2021)  
Diploma thesis: Semisupervised learning for mice recordings analysis.  
Graded with 10/10, supervised by Dr. Th Giannakopoulos (Principal Researcher at NCSR Demokritos), in cooperation with the lab of Prof. Erich Jarvis at Rockefeller University (New York).

## AWARDS, FELLOWSHIPS & DISTINCTIONS

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1. **MIT HEALS Fellowship** **Academic Year 2026-2027**  
Selective, Institute-wide fellowship supporting innovative research with potential impact in health and life sciences
2. **Christos Papakyriakopoulos Award** **2017 & 2018**  
For achieving the highest grade (10/10) in my university class in Mathematics (1<sup>st</sup>, 2<sup>nd</sup>, 3<sup>rd</sup> & 4<sup>th</sup> semester) (Merit-Based Award)

3. **Nikolaos Kritikos Award** **2017**  
Awarded for achieving the highest grade (10/10) in my university class in Mathematics in 1<sup>st</sup> year of studies (Merit-Based Award)
4. **3rd Prize, Hellenic Mathematical Society** **2016**  
Awarded for my performance at the 76<sup>th</sup> Pan-Hellenic (National) Students Contest in Mathematics organized annually by the Hellenic Mathematical Society (Merit-Based Award)
5. **Award from Varvakio Foundation** **2016**  
Awarded for achieving the highest admission grades in Mathematics and Physics at the Pan-Hellenic University Admission Examinations in my School (Merit-Based Award)

## PUBLICATIONS

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1. Petridis, P., Margaritis, G., **Stoumpou, V.**, & Bertsimas, D. (2026). Holistic AI in medicine; improved performance and explainability. NPJ Digital Medicine 9 (1), 120.
2. Bertsimas, D., Everest, L., & **Stoumpou, V.** (2025). Multimodal prescriptive deep learning. Transactions on Machine Learning Research, 2026. ISSN 2835-8856. URL <https://openreview.net/forum?id=AwfWOCVLbJ>.
3. **Stoumpou, V.**, Vargas, C. D., Schade, P. F., Boyd, J. L., Giannakopoulos, T., & Jarvis, E. D. (2023). Analysis of Mouse Vocal Communication (AMVOC): a deep, unsupervised method for rapid detection, analysis and classification of ultrasonic vocalisations. Bioacoustics, 32(2), 199-229.
4. **Stoumpou, V.**, Bertsimas, D., Singer, S., & Margonis, G.A (2026). Observing the unobserved confounding through its effects: toward randomized trial-like estimates from real-world survival data. arXiv preprint arXiv:2604.12137, 2026.
5. **Stoumpou, V.**, Tysarowski, M., Azemi, T., Haider, J., Haronian, H. L., Hagberg, R. C., & Bertsimas, D. (2025). An Interpretable AI Tool for SAVR vs TAVR in Low to Intermediate Risk Patients with Severe Aortic Stenosis. arXiv preprint arXiv:2512.10308.
6. **Stoumpou, V.**, Kumar, R., Burman, B., Ojeda, D., Mehta, T., & Bertsimas, D. (2025). Detection and Localization of Subdural Hematoma Using Deep Learning on Computed Tomography. arXiv preprint arXiv:2512.09393.
7. Paschalidis, P., **Stoumpou, V.**, Everest, L., Ma, Y., Azemi, T., Haider, J., ... & Bertsimas, D. (2026). Towards Optimal Valve Prescription for Transcatheter Aortic Valve Replacement (TAVR) Surgery: A Machine Learning Approach. Health Care Management Science 29 (3), 27.

8. Bertsimas, D., Everest, L., Gu, J., Peroni, M., & **Stoumpou, V.** (2024). Deep Trees for (Un) structured Data: Tractability, Performance, and Interpretability. arXiv preprint arXiv:2410.21595. *(Under revision in Machine Learning)*
9. Bertsimas, D., & **Stoumpou, V.** (2024). Binary Classification: Is Boosting stronger than Bagging?. arXiv preprint arXiv:2410.19200.

## BOOK PROJECT

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**Bertsimas, D. & Stoumpou, V.** *Holistic AI for Diagnostics, Prognostics and Therapeutics in Medicine.* In preparation.

## CONFERENCES

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1. INFORMS Healthcare, xHAIM: Multimodal AI for Improved Performance & Explainability, Raleigh, NC, July 2026 (Invited talk)
2. POMS Conference, Multimodal Prescriptive Deep Learning, Reno, NV, May 2026 (Invited talk)
3. IMSI Advances in Quantitative Medical Care, ROAD 3: Mitigating Observed and Unobserved Confounding in Observational Data, Chicago, IL, February 2026 (Lightning talk)
4. INFORMS Annual Meeting, ROAD 3: Mitigating Observed and Unobserved Confounding in Observational Data, Atlanta, GA, October 2025 (Session chair)
5. INFORMS Annual Meeting, Towards Optimal Valve Prescription for Transcatheter Aortic Valve Replacement (TAVR) Surgery: A Machine Learning Approach, Seattle, WA, October 2024 (Invited talk, Session chair)
6. INFORMS Annual Meeting, Multimodal Prescriptive Deep Learning, Phoenix, AZ, October 2023 (Invited talk)
7. INFORMS Healthcare, Multimodal Prescriptive Deep Learning, Toronto, Canada, July 2023 (Invited talk)

## TEACHING EXPERIENCE

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1. **Instructor, Analytics for a Better World**  
Undergrad course, MIT, Cambridge, MA

**Spring 2026**

2. **Curriculum Development Lead, Universal AI**  
MIT Open Learning, MIT, Cambridge, MA **Spring 2025 – Fall 2025**  
Developed recitations, assignments, learning objectives, and other instructional content across 16 AI modules in collaboration with MIT faculty.
3. **Teaching Assistant, Machine Learning under a Modern Optimization Lens**  
Grad course, MIT, Cambridge, MA **Fall 2023 & 2024**
4. **Teaching Assistant, Analytics Edge**  
Executive MBA course, MIT, Cambridge, MA **Spring 2024**
5. **Instructor, Computing in Optimization and Statistics**  
Grad IAP course, MIT, Cambridge, MA **Winter 2025**
6. **Teaching Assistant, Computing in Optimization and Statistics**  
Grad IAP course, MIT, Cambridge, MA **Winter 2024**
7. **Teaching Assistant, Software Analytics Tools**  
Grad orientation class **Fall 2024 & 2026**

## WORK EXPERIENCE

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- **Motor RnD, Tesla, Greece** 03/2022-07/2022  
Worked as an Associate Electrical Design Engineer.  
Main tasks: App development, Software engineering, Thermal model validation for motors.

## UNDERGRADUATE RESEARCH EXPERIENCE

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- Participation in SURE (Summer Undergraduate Research Experience) program at the **University of Southern California, Los Angeles (Department of Electrical & Computer Engineering)**. Worked in the Neural Systems Engineering & Information Processing Lab with Prof. Maryam Shanechi on issues related to Brain Machine Interfaces. Fellowship awarded by USC (summer 2018).
- Internship at the **Department of Mechanical and Process Engineering, ETH (Zurich)**. Worked for one month in Multiscale Robotics Lab (Prof. D. Nelson group) on issues related to magnetic particle flow characterization. Fellowship awarded by ETH (summer 2019).

## LANGUAGE SKILLS

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- **Greek:** Native Language Proficiency
- **English:** Full Professional Proficiency
- **German:** Full Professional Proficiency